

REINFORCEMENT LEARNING-BASED GENERATIVE DESIGN METHODOLOGY FOR KINETIC FACADE

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Abstract. This paper presents a reinforcement learning (RL) based design method for kinetic facades to optimize the movement direction of shading panels. Included with this research is a case study on the Westin Peachtree Plaza in Atlanta, USA to examine the effectiveness of the proposed design method in a real-life context. Optimization of building performance has been given increased attention due to the significant impact buildings have on energy consumption and carbon emissions. Further, building performance is closely related to the “Sustainable Cities and Communities” mentioned in SDG11. Results show that the novel design method improved the building performance by reducing solar radiation and glare and illustrate the potential of RL in tackling complex design problems in the architectural field.

Keywords. Reinforcement Learning; Kinetic Facade; Generative Design; Design Methodology; SDG 11.

1. Introduction

With the increasing demand for building energy performance and occupant comfort, kinetic facades (building facades with kinetic elements) have attracted increased interest in recent years due to its ability to adapt to the external environment by actively changing itself. For instance, Al Bahr Tower in Abu Dhabi has a kinetic facade system that reduces 50% energy consumption for office spaces and 20% for the whole building (Karanouh and Kerber, 2015).

Many studies show that an optimized control system can improve the performance of kinetic facades (Loonen et al., 2013; Zhang et al., 2018; Smith and Lasch, 2016). However, expensive sensors and actuators in control systems significantly increase the facade cost (Attia et al., 2018). Improving the performance of kinetic facades through generative design creates higher requirements for the design method (Hosseini et al., 2019; Mahmoud and Elghazi, 2016).

The rapid development of artificial intelligence provides a new possibility for

optimizing kinetic facades. In the last 20 years, machine learning achieved considerable growth as a result of the rapid increase of computing power and the availability of large data sets (Russell and Norvig, 2016). Based on a reinforcement learning (RL) algorithm in machine learning, this paper presents a generative design method for kinetic facades. It improves the facade performance by optimizing the movement direction of each unit. There are two design objectives in this research: (1) Reducing the direct solar radiation heat on the building surface; and (2) improving occupants' visual comfort by reducing glare.

The following sections include the introduction of RL, including its theory, algorithms, and applications in the architectural field. Then the novel design system will be described based on the RL framework. A case study on the Westin Peachtree Plaza in Atlanta, Georgia is included to show a precedent for the design process alongside an experiment to verify the effectiveness of this design method. The last two sections discuss the experiment results and the contribution of this research.

2. Reinforcement Learning in Design

2.1. REINFORCEMENT LEARNING

Machine learning algorithms have multiple classes of learning: Supervised, Unsupervised, Semi-supervised, and Reinforced. The machine learning algorithm used in this study is RL. Compared with other algorithms, RL increases the efficiency of solving complex design problems with no training set and model-free algorithms (Sutton and Barto, 2018). Figure 1 shows the framework of the RL system.



Figure 1. Framework of Reinforcement Learning

The two indispensable subjects in this system are agent and environment. The agent can observe and implement action to the environment. After that, the environment can change its state based on the agent's action and give feedback to the agent. The agent learns how to maximize rewards through continuous interaction with the environment. In addition, the specific algorithm determines the way the agent learns from interactions.

2.2. Q-LEARNING ALGORITHM

The specific algorithm used in this research is Q-learning. Q-learning is a widely used model-free RL algorithm based on Markov Decision Process (MDP). Model-free MDP algorithms include Monte-Carlo Learning and Temporal-Difference Learning. Q learning uses the Q table to evaluate the pros and cons of different actions in a state. The learning process of Q-learning is the updating process of the Q table. Based on the Bellman equation, the updating method of the Q table in this research could be

described by the following equation:

$$\text{New}Q(s, a) = (1 - \alpha)Q(s, a) + \alpha[R(s, a) + \gamma \max_{a'} Q(s', a')] \quad (1)$$

In this equation, $Q(s, a)$ and $\text{New}Q(s, a)$ represent the current and the updated Q value. α is the learning rate of the model. A higher learning rate means less impact from former training. $R(s, a)$ is the reward for taking that action in that state where γ represents the discount rate. $\max_{a'} Q(s', a')$ is the maximum expected Q value for the next state. With a lower γ value, the agent cares more about the reward in the current step rather than the expected future rewards.

2.3. APPLICATIONS IN DESIGN

Markov Decision Process (MDP) is the most common model in RL. The future state of the environment is only related to the present in MDP. It is often necessary to assume the environment and the agent follow the MDP model when applying RL in the design field due to the complexity of real-world design issues. The application of RL in the architectural field mainly focused on intelligent building control strategy and design generation. For instance, Zhang et al developed a Deep RL-based HVAC control framework, and the experiment result shows that this system reduces 15% of heating energy (Zhang et al., 2018). In addition, Smith and Lasch used RL in controlling adaptive facade systems through an Intelligent Adaptive Control framework (Smith and Lasch, 2016).

In the direction of design generation, Chang et al developed a design method that combines RL, parametric object modeling, and multivariate statistical approach to get better energy performance (Chang et al., 2019). Han et al proposed a design method for urban blocks combining deep RL and computer vision (Han et al., 2020). This method could optimize the performance and aesthetics of urban blocks. Additionally, Veloso and Krishnamurti generated spatial configurations to address specific spatial goals based on double deep Q-network (DDQN) and dynamic convolutional neural-network (DCNN) (Veloso and Krishnamurti, 2020).

3. Reinforcement Learning-Based Generative Design Method

3.1. DESIGN SYSTEM

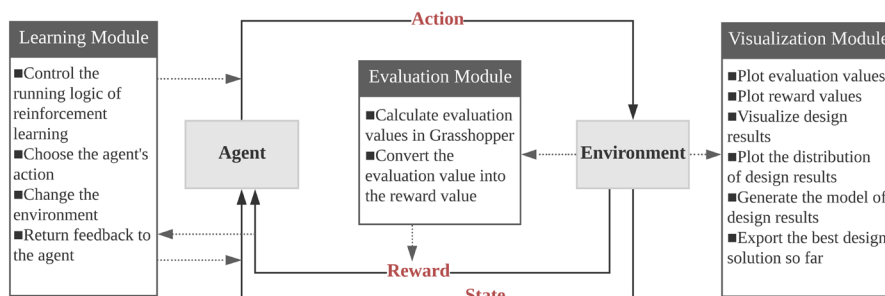


Figure 2. Design system framework

Based on the Agent-Environment framework of RL, this research proposed a design system as shown in Figure 2. Following the principle of modularity, this design system could be decomposed into three modules: Learning, Evaluation, and Visualization.

Learning module is the core of the design system and controls the logic of the RL process. Since the RL algorithm is Q-learning in this research, the learning module constructs a Q table based on the design issue. Further, the learning module chooses the agent's action and gives the corresponding feedback to the agent. After that, the learning module changes the environment based on the agent's actions.

Based on the optimization direction of this design method, the evaluation module gets the environment info from Python code and then inputs data into Grasshopper to calculate the reward reference value for corresponding optimization directions. After that, the evaluation module converts the reference value to the reward value.

The former two modules already form a workable RL system. This design system also constructed a visualization module to monitor the training process and present the design results. This module plots the change of evaluation values during the training process. In addition, this module visualizes the design results and prints the design solution with the best performance so far. It will help the designer to monitor the training process and select the final design solution.

3.2. DESIGN PROCESS

Figure 3 shows the detailed design process. Using Westin Peachtree Plaza as an example, the following sections describe the process from analyzing design requirements to getting the design solution.

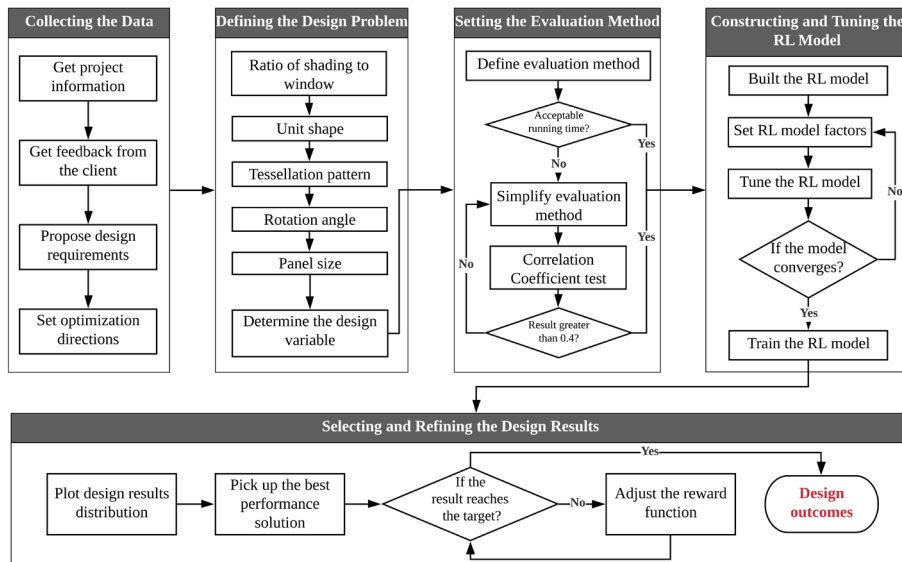


Figure 3. Design process of the RL-based design method

3.2.1. [Step 1]: Collecting the Data

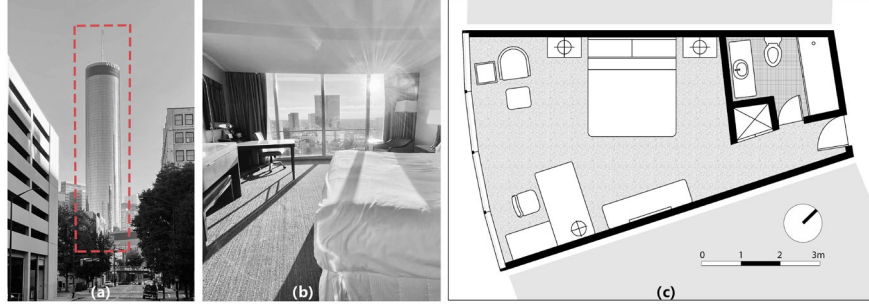


Figure 4. Project information. (a): Westin Peachtree Plaza (image taken by the author), (b): Glare in the guest room (image taken by the author), (c): Guest room plan

The first step in this design method is to collect the project information and propose design requirements. It provides a basis for the following optimization. This example is based on a guest room in Westin Peachtree Plaza which is oriented 35° west of south. It can be seen from Figure 4 that this room has a serious glare in the afternoon. So the design requirements in this example are less solar radiation and glare. Considering the shading efficiency and daylighting requirements, the kinetic facade system in this research adopts the operation mode shown in Figure 5.

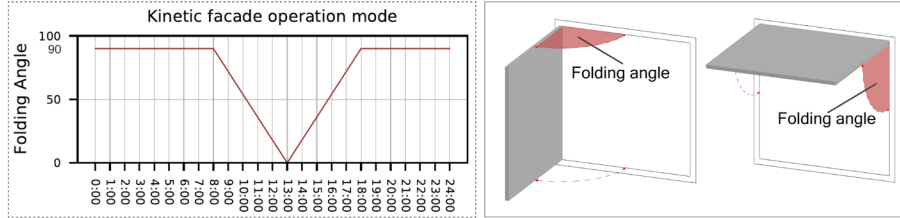


Figure 5. Kinetic facade operation mode

3.2.2. [Step 2]: Defining the Design Problem

Considering the variety of design decisions in kinetic facades design, it is hard to make all the design decisions based on RL. The design method needs to further specify the design problem for RL. As shown in Figure 6, the designer needs to make decisions from setting the covering rate of the facade panel to designing the facade pattern. Considering the form of the overall building and shading efficiency of the kinetic facade, the designer defines a kinetic facade pattern with a 100% shade to window ratio and 32 square panels. After that, the design problem in this example is how to determine the folding direction of each panel.

3.2.3. [Step 3]: Setting the Evaluation Method

Based on optimization directions of reducing the solar radiation and glare, this step needs to propose a corresponding evaluation method for these two optimization directions. The evaluation method is the basis of reward calculation, so this step is the

core step in this design method that bridges the building performance and RL. The setting of the evaluation method considers not only the accuracy but also the running time since there are massive calculations in training the RL model. To the consideration of efficiency and accuracy, this design method divides the time from 8 am to 6 pm on the summer solstice (since it has the longest daytime in a year) into ten units (each unit is one hour) and assumes the kinetic facade and solar is static in each time unit, to reduce the calculation amount.

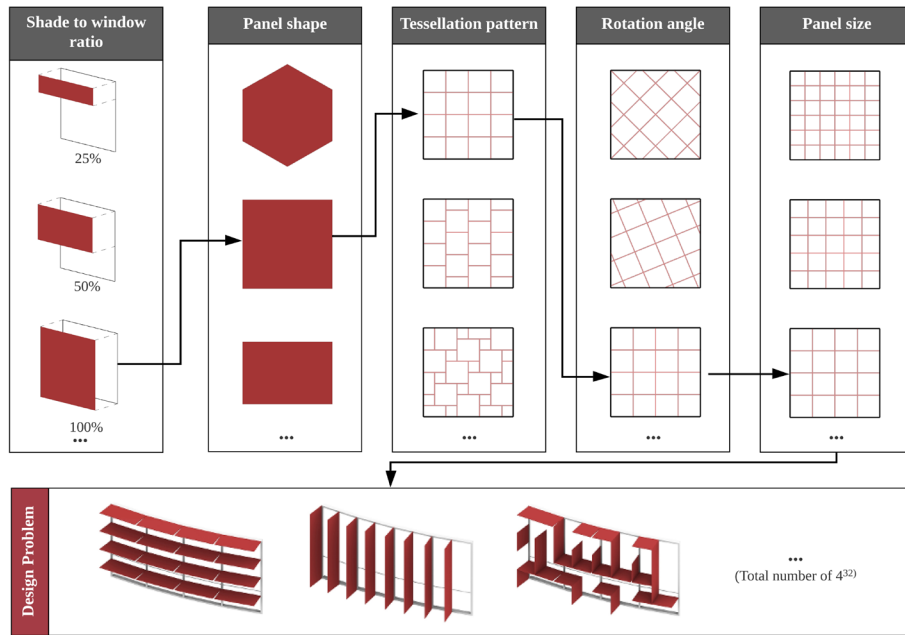


Figure 6. Design process of the facade pattern

For the solar radiation evaluation, the design method firstly calculates the position of shading panels in each time unit based on the operation mode and time (shown in Figure 7). After that, calculate unshaded areas based on the position of the shading panels. Then, project unshaded areas to the plane that is perpendicular to the solar ray and calculate the radiation heat based on the Direct Normal Irradiance (DNI) value.

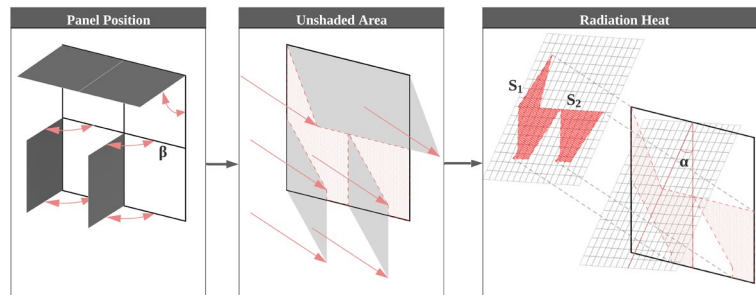


Figure 7. Solar radiation evaluation

For the glare evaluation, there is a widely accepted evaluation method for the glare in buildings called Daylight Glare Probability (DGP). However, the calculation of DGP based on simulation (such as using Honeybee plugin) takes a long time in RL training. Considering that direct sunlight will produce a serious glare in the room, this design method assumes the volume of sunlight in the room is positively correlated to the glare issue. To further test the accuracy of using sunlight volume as the glare evaluation criteria, this research constructed a Pearson Correlation Coefficient test with solar beam volumes and DGP values. This test was based on 100 randomly generated facades and the magnitude of the test in this research is 0.53 which means the solar beam volumes and the DGP values are high-degree correlated.

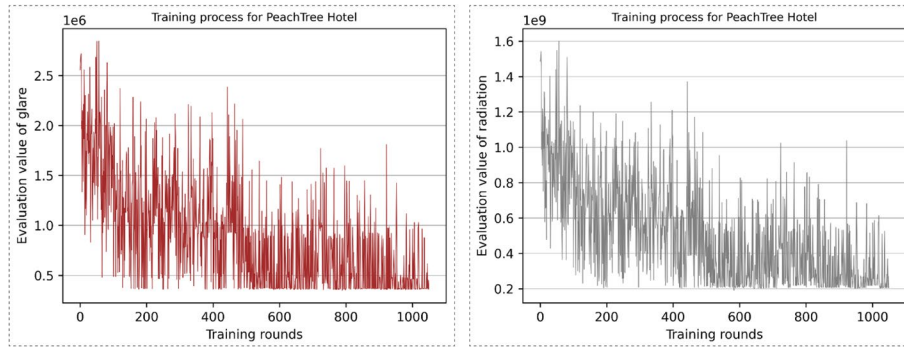


Figure 8. Training process of RL model

3.2.4. [Step 4]: Constructing and Tuning the RL Model

This step gathers all the information in the former steps and converts them into a feasible RL model. It starts from setting the reward calculation equation. This research uses the following equation to calculate the reward in each step. S_n and G_n are changes of radiation and glare evaluation values in each step, and d and g are factors of solar radiation and glare. For example, the reward will only be related to the glare evaluation value when d is equal to zero. The designer can use these two factors to adjust the preference of reducing solar radiation and glare in the training process.

$$R = d * S_n + g * G_n \quad (2)$$

Then, this step sets the critical parameters of the RL model and trains the RL model after several rounds of tuning. Figure 8 shows the training process of the RL model. It is worth noting that the agent has more stable performance and is more likely to get design results with less radiation and glare evaluation value (means better performance in blocking radiation and glare) with the increasing training rounds.

3.2.5. [Step 5]: Selecting and Refining the Design Results

In 1000 rounds of RL training, the design system generates many design solutions. This step firstly visualizes the distribution of these design results to provide a reference for the designer to select the final design result.

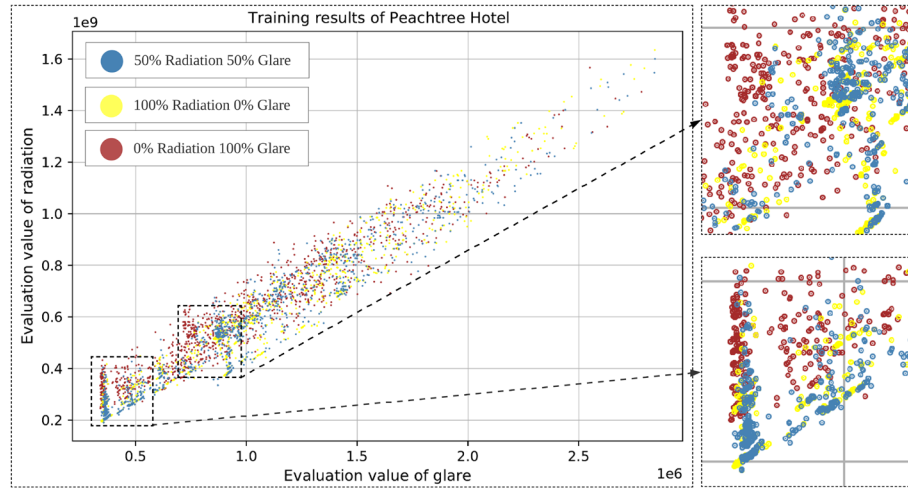


Figure 9. Distribution of design results

By changing the value of d and g in the former equation, the designer can redo the training and get new design results with the same system. For example, Figure 9 shows the distribution of design results with different compositions of rewards. A point closer to the x-axis means better performance in reducing radiation, and a point closer to the y-axis means better performance in blocking glare. In this example, the designer finally selected the design result closest to the origin, and Figure 10 shows the corresponding facade rendering at different times of a day.

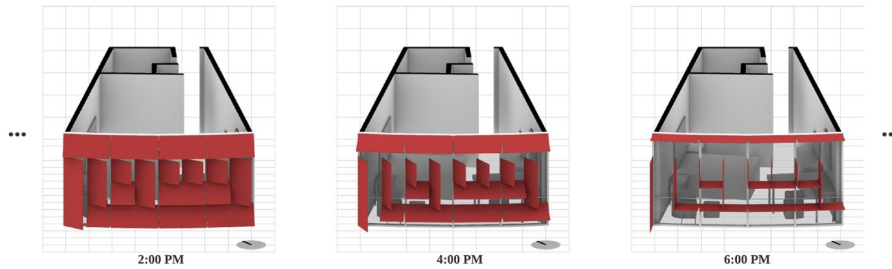


Figure 10. States of the final design result at different times of a day

4. Experiment

To further verify the effectiveness of the RL-based design method, this research proposed an experiment with more sampling data and higher sampling density. This experiment is based on the real data of the Westin Peachtree Plaza guest room and the weather data in Atlanta. The independent variable in this experiment is the kinetic facade design. The dependent variables include the Direct Solar Radiation and Daylight Glare Probability (DGP).

This experiment tested the performance of different kinetic facade designs (include pure horizontal facade, pure vertical facade, RL generated facade, randomly generated

facade, and the original none shading facade as a base line) in 91 days from 5/7 to 8/5 (summer solstice is the midpoint) with a ten minutes time unit. The calculation of the solar radiation experiment is based on geometric calculation functions of Grasshopper in Rhino. In addition, the glare experiment is based on the Honeybee plugin in Grasshopper. Figure 11 shows the results for the solar radiation experiment and glare experiment.

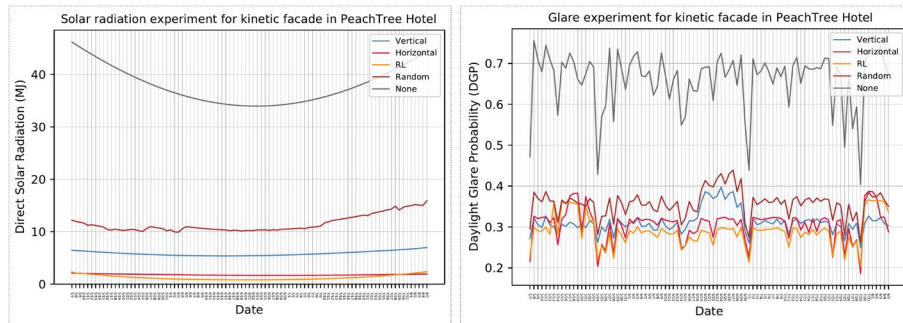


Figure 11. Experiment results for kinetic facades in Westin Peachtree Plaza

5. Discussion

As shown in Figure 12, this study analyzed the distribution of experiment results using boxplots. Further, it investigated the cumulative experiment results using bar charts. It is worth noting that the RL generated facade has the best performance in both the solar radiation and the glare experiment with the hybrid vertical and horizontal panels. Compared with the pure vertical facade, the RL generated facade reduced the cumulative solar radiation by 78.3% and the cumulative DGP by 7.1%. In addition, the RL generated facade reduced the solar radiation by 29.7% and DGP by 5.1% compared with the pure horizontal facade.

The rule of thumb in designing shading devices is that the horizontal louver is good at blocking solar radiation and the vertical fin is good at blocking glare, especially in west orientation. Experiment results show that the pure horizontal kinetic facade performs well in blocking solar radiation and glare in a Southwest orientation. In addition, the pure vertical kinetic facade has good performance in blocking glare but is not so good in blocking solar radiation.

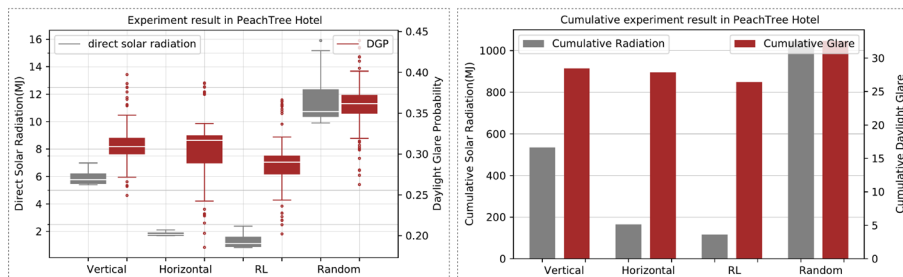


Figure 12. Left: Distribution of experiment results. Right: Cumulative experiment results

6. Contribution

The experiment results show that the kinetic facade generated by the novel design method has better performance than the facades generated by traditional design methods. Future research will further compare this RL-based design method with model-based design methods. Changing states to respond to different environments and design objectives, this method revealed more possibilities for adapting to future challenges reducing life-cycle carbon emissions.

We assert that this method improves the performance of kinetic facades in early design stages which could increase the efficiency of buildings with minimal energy consumption. It will further promote the realization of the "Sustainable Cities and Communities" described in SDG11. In addition, the design method based on Python and Grasshopper could reduce the learning curve for designers, which will help the spread of RL-based design methods. This research expands the field of generative design but also provides reference and inspiration for the application of RL in design.

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